

1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Full Form	Double Data Rate 3	Double Data Rate 4	Double Data Rate 5
Speed Range	800–2133 MT/s	2133–4800 MT/s	4800–8400+ MT/s
Voltage	1.5V (1.35V for DDR3L)	1.2V	1.1V
Power Consumption	Higher	Lower than DDR3	Lowest among the three
Maximum Capacity per Module	Up to 16 GB	Up to 64 GB	Up to 128 GB+
Performance	Good for older systems	Better multitasking and gaming	Best for high-performance computing and gaming
Release Year	2007	2014	2020
Typical Use Cases	Older desktops and laptops, basic computing	Modern PCs, gaming systems, office work	Latest gaming PCs, AI workloads, video editing, workstations
Compatibility	DDR3 motherboards only	DDR4 motherboards only	DDR5 motherboards only
Cost	Cheapest	Moderate	Most expensive

3.List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)	NVMe SSD
Speed	Slowest	Faster than HDD	Fastest
Technology	Uses spinning disks	Uses flash memory	Uses flash memory with PCIe interface
Typical Use	Large file storage and backups	Everyday computing and applications	Gaming, video editing, and high-performance tasks

Difference	HDD	SSD	NVMe
Data Transfer Speed	Low	Medium	Very High
Boot/Application Load Time	Slow	Fast	Extremely Fast
Price per GB	Cheapest	Moderate	Most Expensive

Storage Type	Example App / Use Case	Why It Benefits
HDD	Movie Collection (Videos, Photos, Downloads)	Large storage capacity at low cost
SSD	Spotify / Netflix	Faster app launching and smooth performance
NVMe SSD	FIFA, PUBG PC, Valorant, Adobe Premiere Pro	Very fast loading times and quick data access

